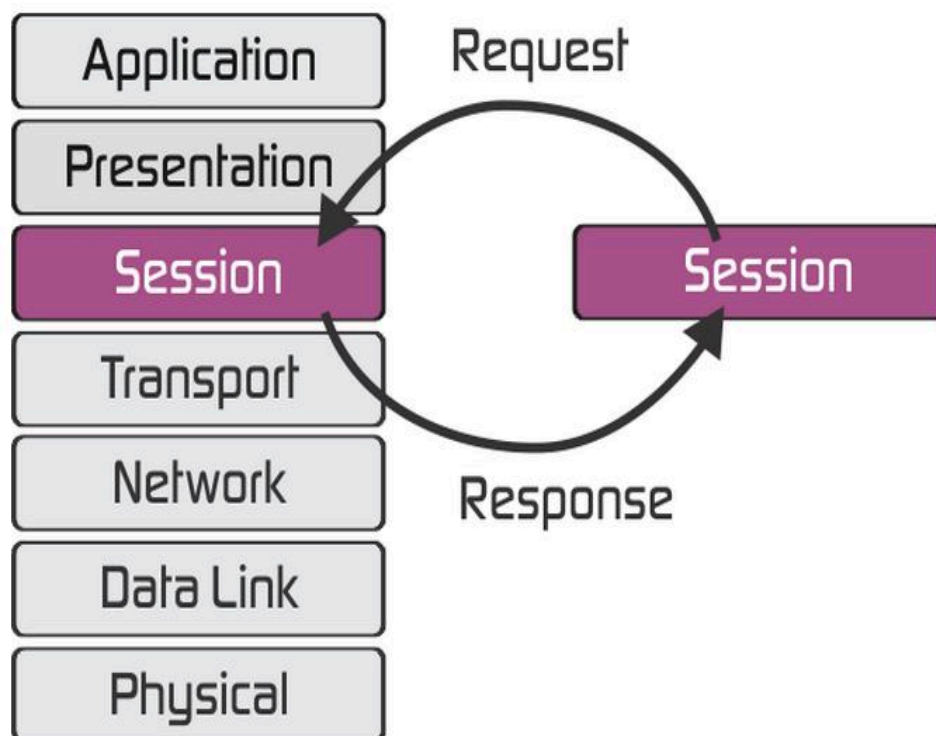
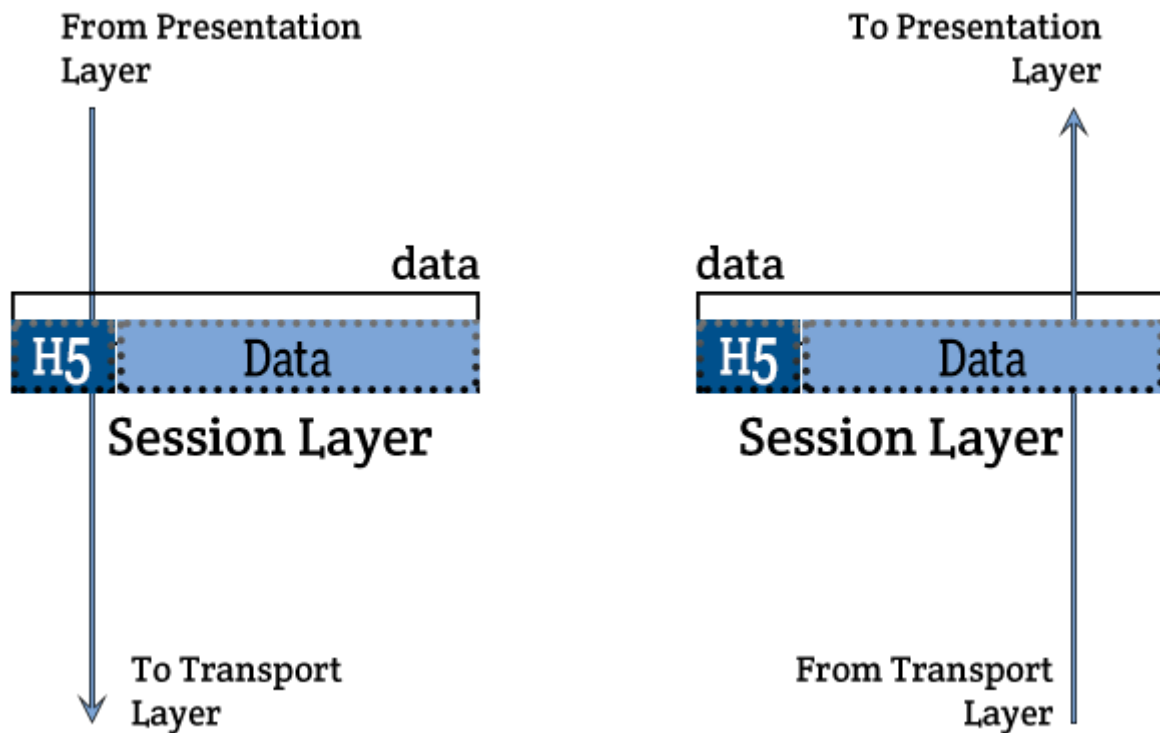

Unit V

Session layer: Overview, Authentication, Session layer protocols, Presentation layer: Overview, Data conversion, Encryption and Decryption, Presentation layer protocols (LPP, Telnet, X.25 packet Assembler/Disassembler), Application Layer: Domain name system, SNMP, Electronic Mail; the World WEB, HTTP, FTP.

Session Layer

The Session Layer is the 5th layer in the Open System Interconnection (OSI) model which plays an important role in controlling the dialogues (connections) between computers. This layer is responsible for setting up, coordinating, and terminating conversations, exchanges, and dialogues between the applications at each end. It establishes, manages, and terminates the connections between the local and remote applications.





- The Session Layer is responsible for establishing active communication sessions between two devices.
- In the OSI model, the transport layer is not responsible for releasing a connection. Instead, the session layer is responsible for that. However, in modern TCP/IP networks, TCP already provides orderly closing of connections at the transport layer.
- Dialogue Control is also implemented in the Session Layer of the OSI model but in TCP/IP the dialogue control is implemented in the Application Layer.
- Session-layer services are commonly used in application environments that use remote procedure calls (RPCs).
- Zone Information Protocol in AppleTalk is an example of Session Layer Implementation.
- Session Layer has synchronization and resynchronization techniques that ensure reliable and orderly communication over networks, which is particularly important in applications requiring high levels of data integrity and continuity.
- Synchronization points are markers or tokens inserted into the data stream that allow communication sessions to have checkpoints and on the other hand Resynchronization involves restoring a session to a known state after a disruption, such as a network failure or session timeout.

Functions of Session Layer

The Session layer is the fifth layer of the OSI (Open System Interconnection) model. The session layer is responsible for maintaining, establishing, synchronizing, and terminating sessions between them. A connection is being established among the session entities by the session layer. The data received from the presentation layer is handled by the session layer.

Various protocols such as PPTP, PAP, RTCP, ADSP, and SDP are used by the session layer.

Functions of Session Layer

The session layer is responsible for various tasks that are mentioned below:

Functions of Session Layer

- | | |
|----------------------------------|----------------------|
| 1. Session Establishment | 4. Dialog Management |
| 2. Communication Synchronization | 5. Data Transfer |
| 3. Activity Management | 6. Resynchronization |

1. Session Establishment

It establishes connection-oriented or connectionless sessions between communicating parties for reliable data transfer (e.g., files, remote login).



Session of Communication

1. Session Establishment

The basic and most important function of the session layer is to establish a connection between the communicating parties known as sessions. This connection helps to transfer the data such as files, remote login, and communication over these sessions in a more ordered and reliable way. The session layer can establish connection-oriented as well as connectionless sessions. Whenever any session is released, the transport connection is also released. Sessions can be mapped onto transport connections in three ways. They are:

- **One-to-one mapping:** In one-to-one mapping, one transport connection is being used by one session.

- **Many-to-one mapping:** In many-to-one mapping, the same transport connection is being used by the consecutive sessions.
- **One-to-many mapping:** In one-to-many mapping, multiple transport connections are used by one session.

2. Communication Synchronization

It adds synchronization checkpoints in long data streams to ensure error recovery and proper data transmission.



The session layer provides proper connectivity among the systems, during this if any occurs, a recovery option known as the state is provided. Using the known state session layer adds synchronization bits in every state of the error and these bits are then used as checkpoints. For longer and complex communications the checkpoints are added to the data stream. It makes sure that data streams are received successfully and acknowledged up to the checkpoints. If any failure occurs then only the stream is retransmitted after the checkpoints. The functionality is being provided by the session layer because the transport layer can mask only the communication errors, not the upper layer errors.

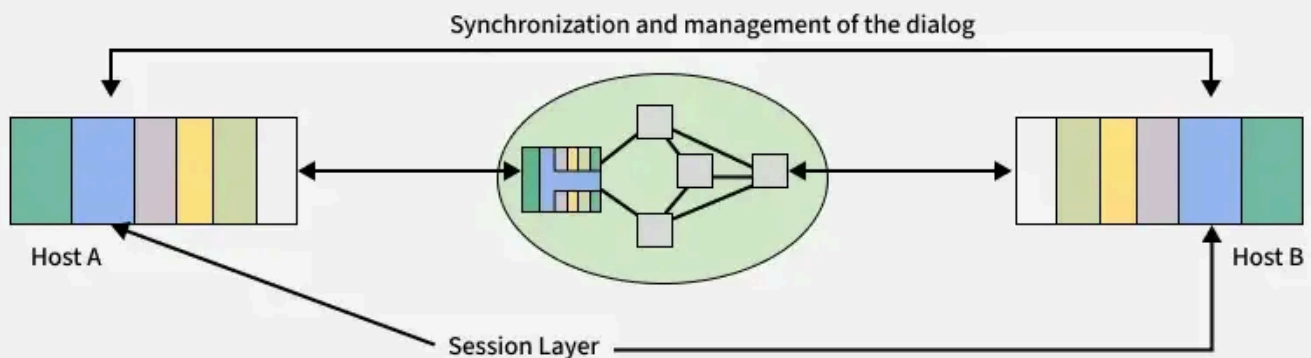
3. Activity Management

It splits message streams into logical units called "activities," allowing differentiation of tasks performed during a session.

The session layer provides the feature for users to differentiate among the activities that are performed during the session. The message stream can split into logical units and these logical units are known as activities. Activity management is defined as a process of splitting of message stream into activities.

4. Dialog Management

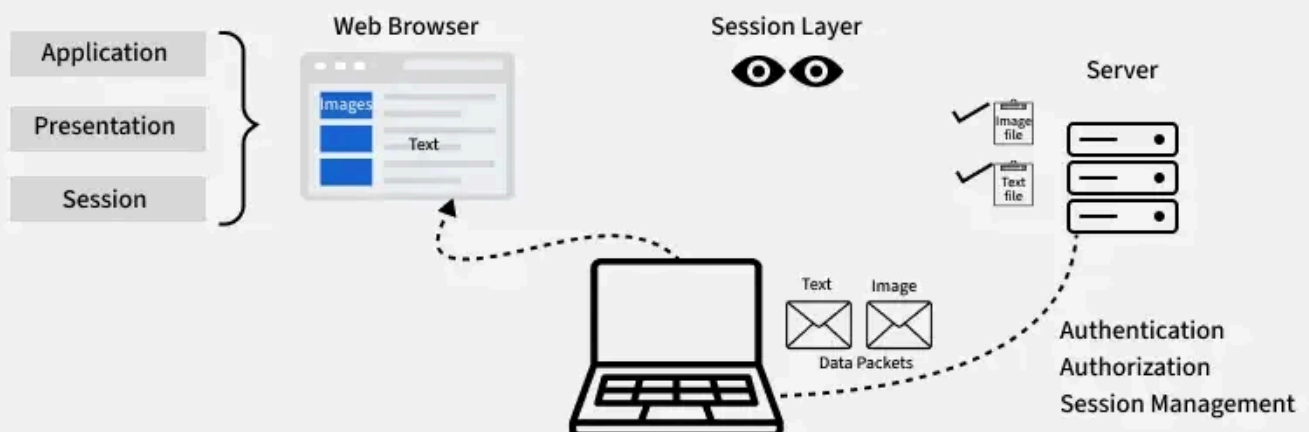
It manages session dialogs, including initiating, maintaining, and ending communication.



Dialog management is defined as a process of keeping log data for the connections established for transmitting and receiving data. It is required for establishing, synchronization, preserving and ending of the conversation between the communicating parties namely sender and the receiver. For this a mechanism known as token mechanism is being used in which a user that is sharing the data is given a token if it is in half duplex mode and after the exchange it is transferred to another device. If the operation is in full duplex mode then no token is used for transmission of data. The efficiency of the connection is maintained if the token mechanism is used.

5. Data Transfer

It handles data exchange between users in; ensuring authentication and authorization.



Data transfer is also one of the basic and important functions of the session layer. Session

layer handles the exchange of data between the communicating systems. This exchange of data between the users can be in two modes of transmission. They are:

- **Half-duplex mode:** In half-duplex mode of transmission of data only one user has the right to initiate the data transfer.
- **Full- duplex mode:** In full duplex mode of transmission data can be transmitted in both the directions simultaneously.

6. Resynchronization

It restores session states to a previously defined synchronization point.

Resynchronisation is also known as backward synchronization. Resynchronisation is defined as a process in which the state of dialog is being restored to its previously defined state. After the resynchronization all the tokens are restored to the positions that were set during synchronization. There are various options of resynchronisation. They are:

- **Set:** Set option is being used for setting the synchronization point serial number to any value that is being chosen by the user.
- **Abandon:** Abandon option is being used for setting the synchronization point serial number to an unused value.
- **Restart:** Restart option is being used for setting the synchronization point serial number to any of the used value that is greater than the synchronization point serial number which identifies the last acknowledged major synchronization point.

Working of Session Layer

- The Session Layer manages communication sessions between applications over a network.
- It establishes connections, negotiating session parameters like authentication and communication direction (full-duplex or half-duplex).
- It oversees data exchange by using tokens to manage transmission rights and prevent collisions.
- Synchronization techniques are implemented, inserting checkpoints for recovery

in case of disruptions.

- It ensures orderly communication, reducing message loss, duplication, or errors caused by overlapping communication.
- The Session Layer gracefully terminates the session, ensuring all data is exchanged and both sides agree to close

Session Layer Protocols

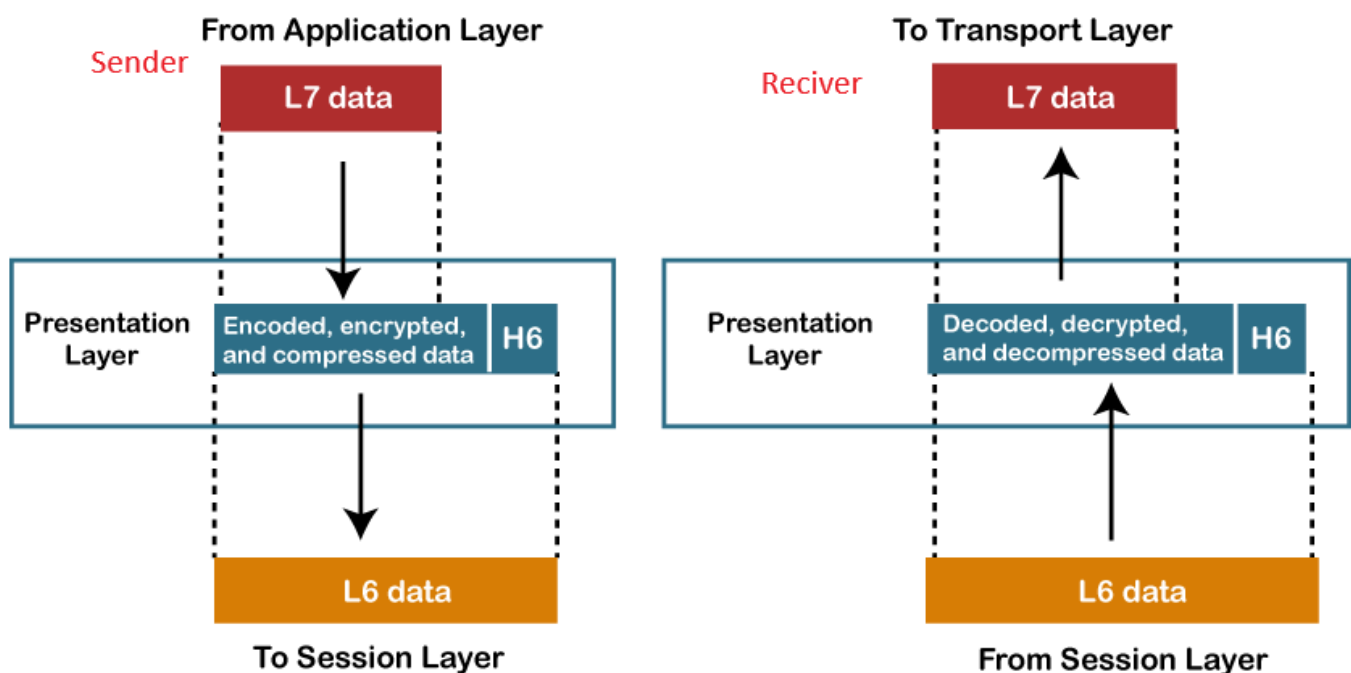
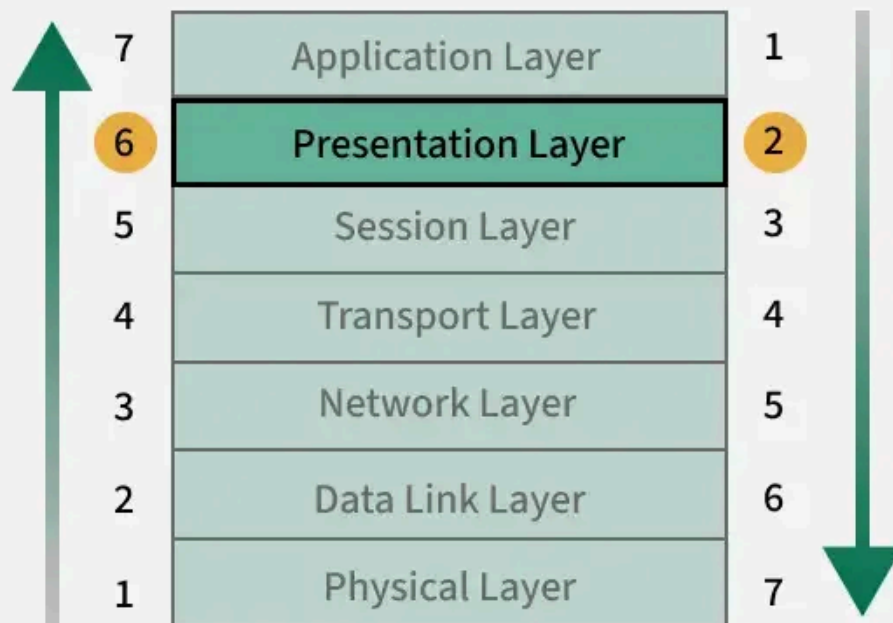
Session Layer uses some protocols which are required for safe, secure and accurate communication which exists between two-end user applications. Following are some of the protocols provided or used by the Session Layer:

- **AppleTalk Data Stream Protocol (ADSP):** ADSP is that type of protocol which was developed by Apple Inc. and it includes a number of features that allow local area networks to be connected with no prior setup. This protocol was released in 1985. This protocol rigorously followed the OSI model of protocol layering. ADSP itself has two protocols named: AppleTalk Address Resolution Protocol (AARP) and Name Binding Protocol (NBP), both aimed at making system self-configuring.
- **Real-time Transport Control Protocol (RTCP):** RTCP is a protocol which provides out-of-band statistics and control information for an RTP (Real-time Transport Protocol) session. RTCP's primary function is to provide feedback on the quality of service (QoS) in media distribution by periodically sending statistical information such as transmitted octet and packet counts or packet loss to the participants in the streaming multimedia session.
- **Point-to-Point Tunneling Protocol (PPTP):** PPTP is a protocol which provides a method for implementing virtual private networks. PPTP uses a TCP control channel and a Generic Routing Encapsulation tunnel to encapsulate PPP (Point-to-Point Protocol) packets. This protocol provides security levels and remote access levels comparable with typical VPN (Virtual Private Network) products.
- **Password Authentication Protocol (PAP):** It is a password-based authentication protocol used by Point to Point Protocol (PPP) to validate users. Almost all network operating systems, remote servers support PAP. PAP authentication is done at the time of the initial link establishment and verifies the identity of the client using a two-way handshake (Client-sends data and server in return sends Authentication-ACK (Acknowledgement) after the data sent by client is verified completely).
- **Remote Procedure Call Protocol (RPCP):** It is a protocol that is used when a computer program causes a procedure (or a sub-routine) to execute in a different address space without the programmer explicitly coding the details for the remote interaction. This is basically the form of client-server interaction, typically implemented via a request-response message-passing system.
- **Sockets Direct Protocol (SDP):** It is a protocol that supports streams of sockets over Remote Direct Memory Access (RDMA) network fabrics. The purpose of SDP is to provide an RDMA-accelerated alternative to the TCP protocol. The primary goal is to perform one particular thing in such a manner which is transparent to

the application.

Presentation Layer

Presentation Layer is the 6th layer in the Open System Interconnection (OSI) model. This layer is also known as Translation layer, as this layer serves as a data translator for the network. The data which this layer receives from the Application Layer is extracted and manipulated here as per the required format to transmit over the network. The main responsibility of this layer is to provide or define the data format and encryption. The presentation layer is also called as Syntax layer since it is responsible for maintaining the proper syntax of the data which it either receives or transmits to other layer(s).



Functions of Presentation Layer

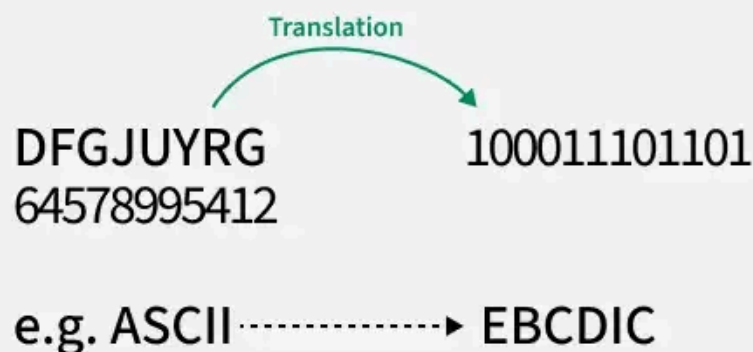
The presentation layer performs several types of functions such as data translation, compression, encryption/decryption etc.

Functions of Presentation Layer

1. Data Translation
2. Data Compression
3. Data Encryption/Decryption
4. Syntax and Semantics Management

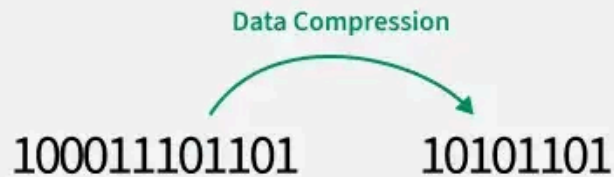
1. Data Translation

It converts data from the application layer into a network-suitable format and translates it back for the receiving application.(e.g., ASCII to EBCDIC)



2. Data Compression

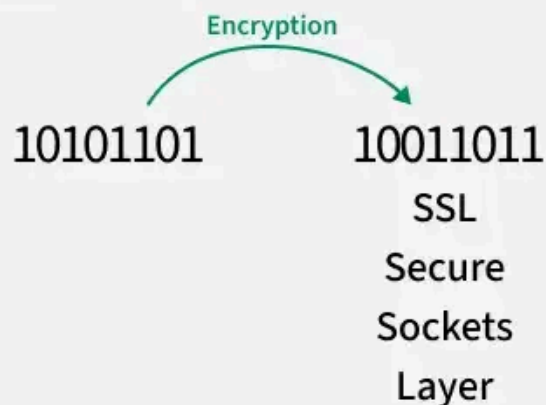
It compresses data to optimize bandwidth and speed, then decompresses it at the receiver's end. Examples: JPEG for images, MPEG for videos, and ZIP for files.



1. Lossy
2. Lossless

3. Data Encryption/Decryption

It secures communication by encrypting data before transmission and decrypting it at the destination. Examples of encryption protocols include SSL and TLS.



4. Syntax and Semantics Management

The presentation layer ensures that the syntax (data structure) and semantics (meaning of data) are properly managed.

Services Provided by Presentation Layer

Presentation layer plays an important role while communication is taking place between two devices in a network.

List of services which are provided by the presentation layer are:

- Presentation layer could apply certain sophisticated compression techniques, so fewer bytes of data are required to represent the information when it is sent over the network.
- If two or more devices are communicating over an encrypted connection, then this presentation layer is responsible for adding encryption on the sender's end as well as the decoding the encryption on the receiver's end so that it can represent the application layer with unencrypted, readable data.
- This layer formats and encrypts data to be sent over a network, providing freedom from compatibility problems.
- This presentation layer is also responsible for compressing data it receives from the application layer before delivering it to the session layer (which is the 5th layer in the OSI model) and thus improves the speed as well as the efficiency of communication by minimizing the amount of the data to be transferred.
- This presentation layer also negotiates the Transfer Syntax.

Presentation Layer Protocols

Presentation Layer, for performing translations or other specified functions, needs to use certain protocols which are defined below –

- **Apple Filing Protocol (AFP)**: Apple Filing Protocol is the proprietary network protocol (communications protocol) that offers services to macOS or the classic macOS. This is basically the network file control protocol specifically designed for Mac-based platforms.

- **Lightweight Presentation Protocol (LPP):** Lightweight Presentation Protocol is that protocol which is used to provide ISO presentation services on the top of TCP/IP based protocol stacks.
- **NetWare Core Protocol (NCP):** NetWare Core Protocol is the network protocol which is used to access file, print, directory, clock synchronization, messaging, remote command execution and other network service functions.
- **Network Data Representation (NDR):** Network Data Representation is basically the implementation of the presentation layer in the OSI model, which provides or defines various primitive data types, constructed data types and also several types of data representations.
- **External Data Representation (XDR):** External Data Representation (XDR) is the standard for the description and encoding of data. It is useful for transferring data between computer architectures and has been used to communicate data between very diverse machines. Converting from local representation to XDR is called encoding, whereas converting XDR into local representation is called decoding.
- **Secure Socket Layer (SSL):** The Secure Socket Layer protocol provides security to the data that is being transferred between the web browser and the server. SSL encrypts the link between a web server and a browser, which ensures that all data passed between them remains private and free from attacks.

Application Layer

The application layer is the top layer of the Open Systems Interconnection (OSI) model, providing an interface between you and a web application. This layer connects software applications and user experiences, allowing you to access network services and data, such as when you check your email or browse the web. The application layer also performs critical processes, such as data conversion, management, and encryption, to communicate between applications.

Functions of the application layer

The application layer facilitates many types of end-user interactions with digital systems. Essential functions of the application layer include:

- Ensuring data transfer integrity and identifying receiving devices for transfer.
- Sending and receiving information, such as databases and files
- Providing communication services, such as email and live chats
- Allowing remote hosting
- Protocol and data syntax determination

Application layer

Several application layer protocols exist that allow devices to connect. Protocols help complete user demands by using a network's program languages. Depending on the task,

you may choose to use one of the following types of application layer protocols:

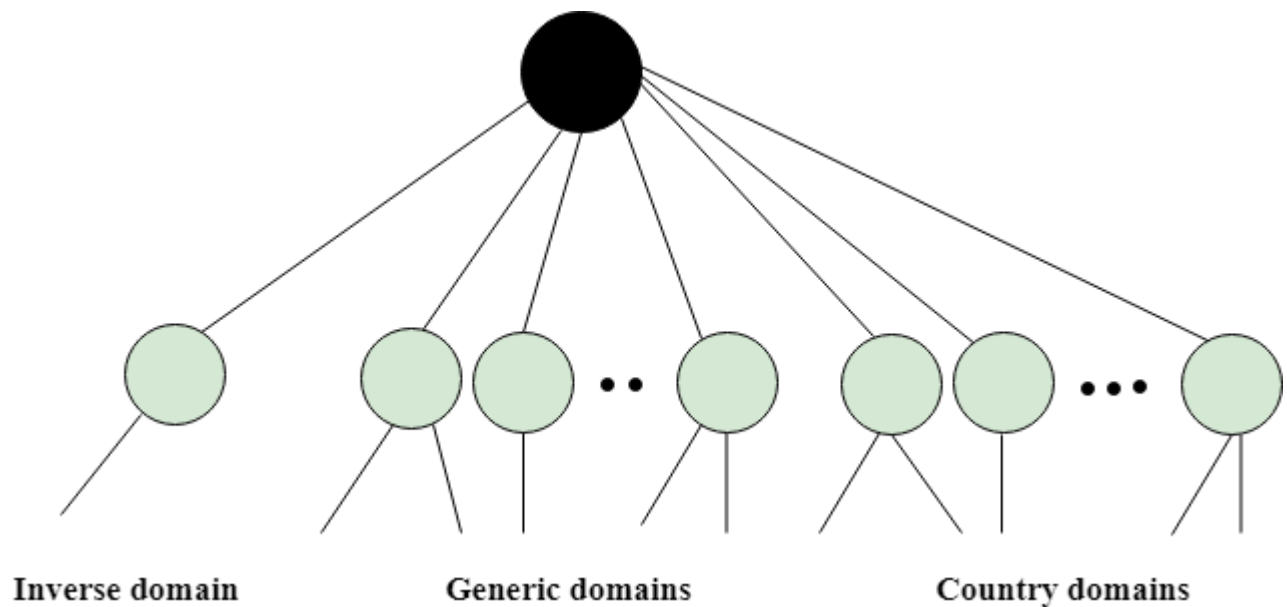
- **File transfer protocol (FTP):** You use FTP to share and transfer files between users and networks.
- **Simple mail transfer protocol (SMTP):** You use SMTP to send and receive email messages. Applications such as Gmail and Outlook use this protocol.
- **Hypertext transfer protocol (HTTP):** You use HTTP to transfer data across the web, including media and web data, by defining commands to request and receive web content.
- **Domain name system (DNS):** You use DNS to translate domain names to IP addresses that can load internet content.
- **Teletype network (TELNET):** You use TELNET to connect with remote devices and systems and relay terminal-to-terminal communication.

DNS

An application layer protocol defines how the application processes running on different systems, pass the messages to each other.

- DNS stands for Domain Name System.
- DNS is a directory service that provides a mapping between the name of a host on the network and its numerical address.
- DNS is required for the functioning of the internet.
- Each node in a tree has a domain name, and a full domain name is a sequence of symbols specified by dots.
- DNS is a service that translates the domain name into IP addresses. This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses.
- For example, suppose the FTP site at EduSoft had an IP address of 132.147.165.50, most people would reach this site by specifying ftp.EduSoft.com. Therefore, the domain name is more reliable than IP address.

DNS is a TCP/IP protocol used on different platforms. The domain name space is divided into three different sections: generic domains, country domains, and inverse domain.

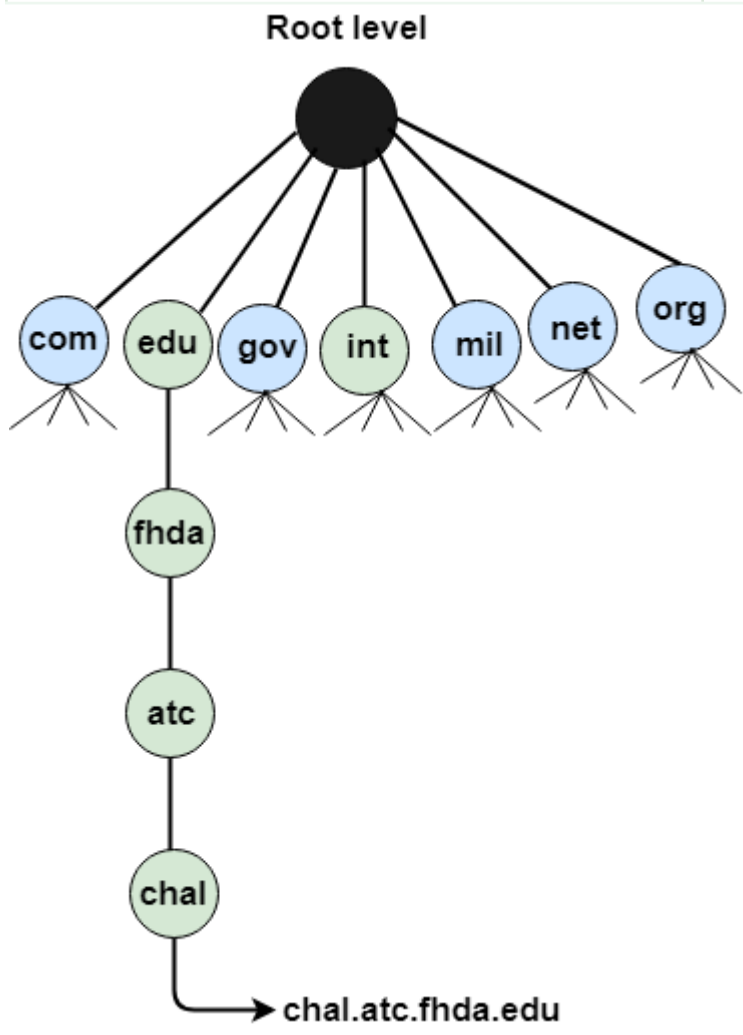


Generic Domains

- It defines the registered hosts according to their generic behavior.
- Each node in a tree defines the domain name, which is an index to the DNS database.
- It uses three-character labels, and these labels describe the organization type.

Label	Description
aero	Airlines and aerospace companies
biz	Businesses or firms
com	Commercial Organizations
coop	Cooperative business Organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International Organizations
mil	Military groups
museum	Museum & other nonprofit organizations
name	Personal names

net	Network Support centers
org	Nonprofit Organizations
pro	Professional individual Organizations



Country Domain

The format of country domain is the same as a generic domain, but it uses two-character country abbreviations (e.g., us for the United States) in place of three character organizational abbreviations.

Inverse Domain

The inverse domain is used for mapping an address to a name. When the server has received a request from the client, and the server contains the files of only authorized clients. To determine whether the client is on the authorized list or not, it sends a query to the DNS server and asks for mapping an address to the name.

Working of DNS

- DNS is a client/server network communication protocol. DNS clients send requests to the server while DNS servers send responses to the client.
- Client requests contain a name which is converted into an IP address known as a forward DNS lookups while requests containing an IP address which is converted into

a name known as reverse DNS lookups.

- DNS implements a distributed database to store the name of all the hosts available on the internet.
- If a client like a web browser sends a request containing a hostname, then a piece of software such as DNS resolver sends a request to the DNS server to obtain the IP address of a hostname. If DNS server does not contain the IP address associated with a hostname, then it forwards the request to another DNS server. If IP address has arrived at the resolver, which in turn completes the request over the internet protocol.

Types of DNS servers

From the outset, developers designed the DNS with a hierarchical, distributed database structure to facilitate a more dynamic approach to domain name resolution, one that can keep pace with a rapidly expanding network of computers. The hierarchy starts with the root level, denoted by a dot (.), and branches out into top-level domains (TLDs)—like “.com,” “.org,” “.net,” or country code TLDs (ccTLDs) like “.uk” and “.jp,”—and second-level domains. The DNS architecture consists of two types of DNS servers, recursive servers and authoritative servers. Recursive DNS servers are the ones doing the “asking,” searching for the information that connects a user to a website.

Recursive servers

Recursive servers—also known as recursive resolvers or DNS resolvers—are typically managed by internet service providers (ISPs), large enterprises or other third-party DNS service providers. They act on behalf of the end user to resolve the domain name into an IP address. Recursive resolvers also cache the answers to a request for a certain period (defined by the time-to-live, or TTL, value) to improve system efficiency for future queries to the same domain.

When a user types a web address into a search browser, the browser connects to a recursive DNS server to resolve the request. If the recursive server happens to have the answer cached, it can connect the user and complete the request. Otherwise, the recursive resolver queries a series of authoritative DNS servers to find the IP address and connect the user to the wanted website.

Authoritative servers

Authoritative servers provide the “answers.” Authoritative name servers hold the definitive records for a domain and respond to requests about domain names stored within their respective zones (typically with answers configured by the domain owner). There are different types of authoritative name servers, each serving a specific function within the DNS hierarchy.

Authoritative DNS name servers include:

Root name servers

Root name servers sit at the top of the DNS hierarchy and are responsible for managing the root zone (the central database for DNS). They answer queries for records stored within the root zone and refer requests to the appropriate TLD name server.

There are 13 IP addresses used to query 13 different root server networks—identified by

letters A through M—that handle requests for the TLDs and direct the queries to the appropriate TLD name servers. The Internet Corporation for Assigned Names and Numbers (ICANN) operates these root server networks.

Top-level domain (TLD) name servers

TLD servers are responsible for managing the next level of the hierarchy, including generic top-level domains (gTLDs). TLD name servers direct queries to the authoritative name servers for the specific domains within their TLD. So the TLD name server for ".com" would direct domains ending in ".com", the TLD name server for ".gov" would direct domains ending in ".gov", and so on.

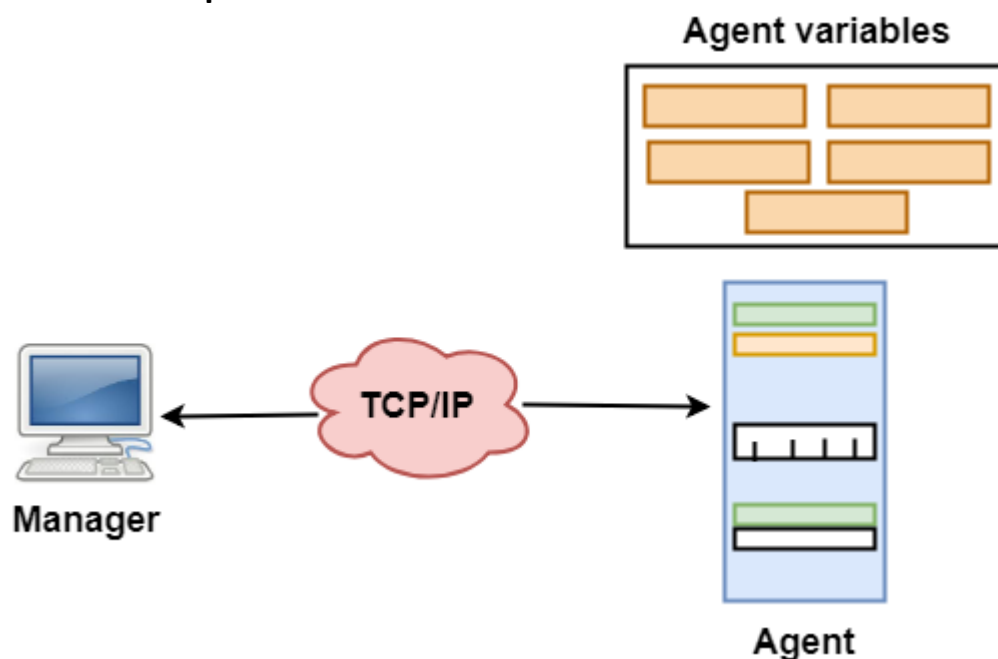
Domain name servers (second-level domain name servers)

The domain name server (sometimes referred to as the second-level domain name server) holds the zone file with the IP address for the full domain name, such as "ibm.com."

SNMP

- SNMP stands for Simple Network Management Protocol.
- SNMP is a framework used for managing devices on the internet.
- It provides a set of operations for monitoring and managing the internet.

SNMP Concept



- NMP has two components: Manager and agent.
- The manager is a host that controls and monitors a set of agents such as routers.
- It is an application layer protocol in which a few manager stations can handle a set of agents.
- The protocol designed at the application level can monitor the devices made by different manufacturers and installed on different physical networks.
- It is used in a heterogeneous network made of different LANs and WANs connected by routers or gateways.

Managers & Agents

- A manager is a host that runs the SNMP client program while the agent is a router that runs the SNMP server program.

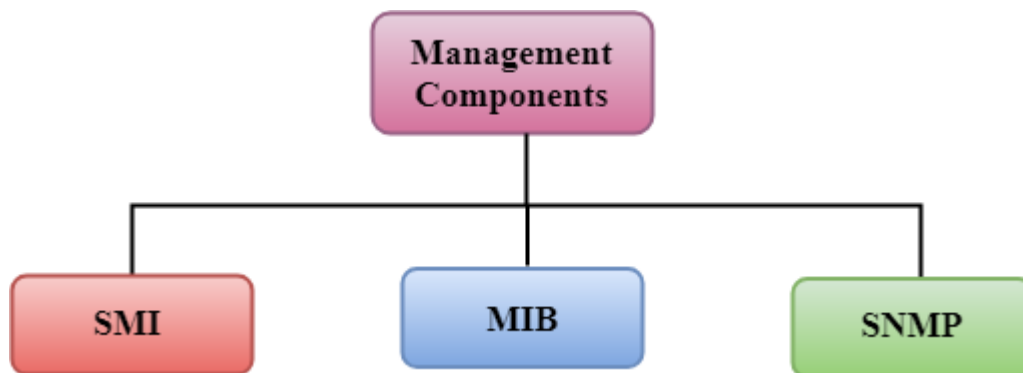
- Management of the internet is achieved through simple interaction between a manager and agent.
- The agent is used to keep the information in a database while the manager is used to access the values in the database. For example, a router can store the appropriate variables such as a number of packets received and forwarded while the manager can compare these variables to determine whether the router is congested or not.
- Agents can also contribute to the management process. A server program on the agent checks the environment, if something goes wrong, the agent sends a warning message to the manager.

Management with SNMP has three basic ideas:

- A manager checks the agent by requesting the information that reflects the behavior of the agent.
- A manager also forces the agent to perform a certain function by resetting values in the agent database.
- An agent also contributes to the management process by warning the manager regarding an unusual condition.

Management Components

- Management is not achieved only through the SNMP protocol but also the use of other protocols that can cooperate with the SNMP protocol. Management is achieved through the use of the other two protocols: SMI (Structure of management information) and MIB(management information base).
- Management is a combination of SMI, MIB, and SNMP. All these three protocols such as abstract syntax notation 1 (ASN.1) and basic encoding rules (BER).

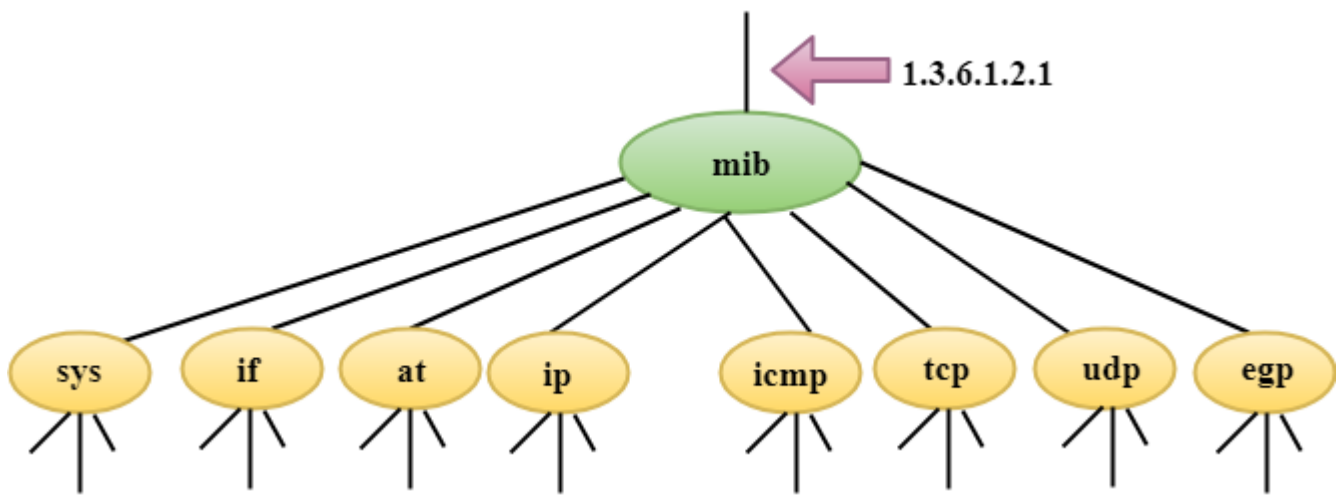


SMI

The SMI (Structure of management information) is a component used in network management. Its main function is to define the type of data that can be stored in an object and to show how to encode the data for the transmission over a network.

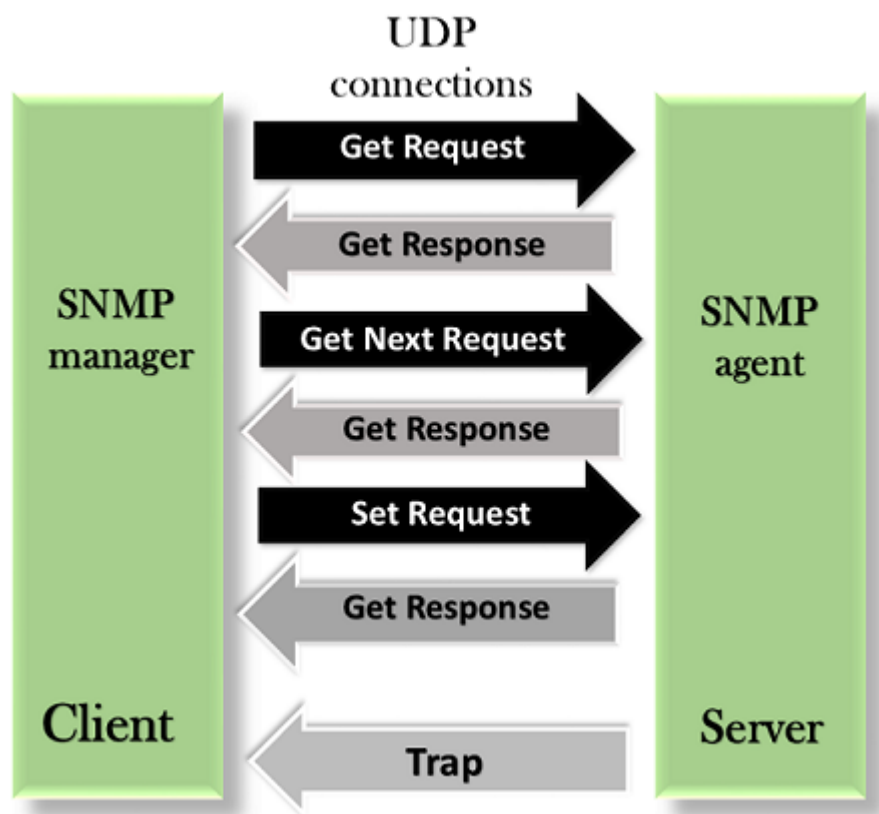
MIB

- The MIB (Management information base) is a second component for network management.
- Each agent has its own MIB, which is a collection of all the objects that the manager can manage. MIB is categorized into eight groups: system, interface, address translation, ip, icmp, tcp, udp, and egp. These groups are under the mib object.



SNMP

SNMP defines five types of messages: GetRequest, GetNextRequest, SetRequest, GetResponse, and Trap.



GetRequest: The GetRequest message is sent from a manager (client) to the agent (server) to retrieve the value of a variable.

GetNextRequest: The GetNextRequest message is sent from the manager to agent to retrieve the value of a variable. This type of message is used to retrieve the values of the entries in a table. If the manager does not know the indexes of the entries, then it will not be able to retrieve the values. In such situations, the GetNextRequest message is used to define an object.

GetResponse: The GetResponse message is sent from an agent to the manager in response to the GetRequest and GetNextRequest message. This message contains the value of a variable requested by the manager.

SetRequest: The SetRequest message is sent from a manager to the agent to set a value in a variable.

Trap: The Trap message is sent from an agent to the manager to report an event. For example, if the agent is rebooted, then it informs the manager as well as sends the time of rebooting.

Electronic mail (e-mail)

Electronic mail is computer networks one of the most well-known network services. Electronic mail is a computer-based service that allows users to communicate with one another by exchanging messages. Email information is transmitted via email servers and uses a variety of TCP/IP protocols. For example, the simple mail transfer protocol (SMTP) is a protocol that is used to send messages. Similarly, IMAP or POP receives messages from a mail server.

- **Spontaneity:** In a couple of seconds, you may send a message to anybody on the globe.
- **Asynchronous:** You may send the e-mail and let the recipient view it at their leisure.
- Attachments of data, pictures, or music, frequently in compressed forms, can be delivered as an e-mail to a person anywhere in the world.
- Addresses can be stored in an address book and retrieved instantly.
- Through an e-mail, a user can transfer multiple copies of a message to various individuals.

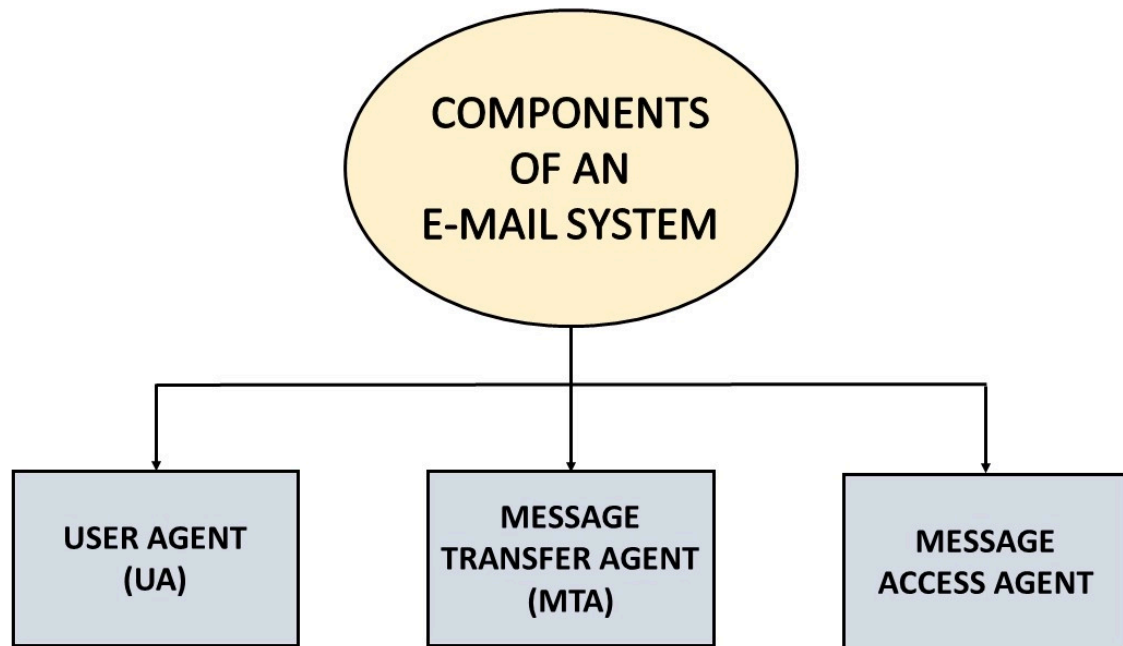
Services offered by Electronic Mail

- **Composition:** Creating messages and responses is referred to as composition.
- **Transfer:** Sending mail from the sender to the receiver is known as a transfer.
- **Reporting:** Mail delivery confirmation is known as reporting. It allows users to see if their mail has been delivered, misplaced, or rejected.
- **Displaying:** It refers to presenting messages so that the user can understand them.
- **Disposition:** This stage concerns the recipient's actions after receiving mail, such as saving it, deleting it before reading it, or after reading it.

Components Of Electronic Mail

The following are the essential components of an e-mail system:

1. User Agent (UA)
2. Message Transfer Agent (MTA)
3. Message Access Agent



User Agent (UA)

The User-Agent is a simple software that sends and receives mail. It is also known as a mail reader. It supports a wide range of instructions for sending, receiving, and replying to messages and manipulating mailboxes.

Some of the services supplied by the User-Agent are listed below:

- Reading a Message
- Sending a reply to a Message
- Message Composition
- Forwarding a Message
- Handling the Message

Message Transfer Agent

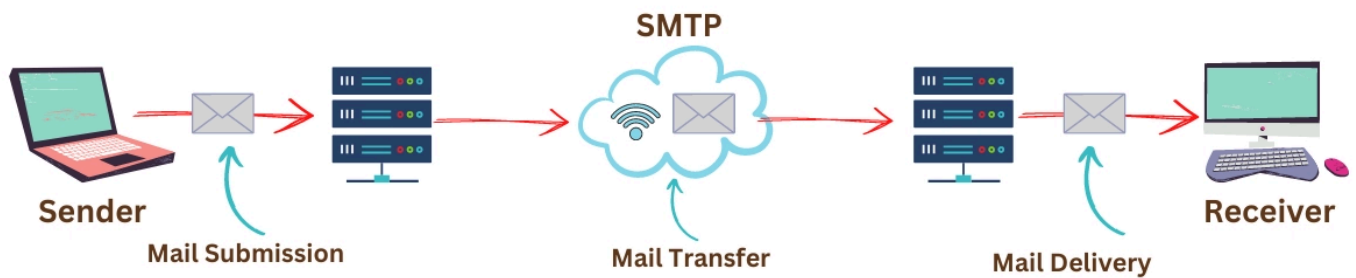
The Message Transfer Agent manages the actual e-mail transfer operation (MTA). Simple Mail Transfer Protocol sends messages from one MTA to another. A system must have a client MTA and a system MTA to send an e-mail. If the recipients are connected to the same computer, it sends mail to their mailboxes. If the destination mailbox is on another computer, it sends mail to the receiver's MTA.

Message Access Agent

The Simple Mail Transfer Protocol is used for the first and second stages of e-mail delivery. The pull protocol is mainly required at the third stage of e-mail delivery, and the message access agent is used at this point.

POP and IMAP4 are the two protocols used to access messages.

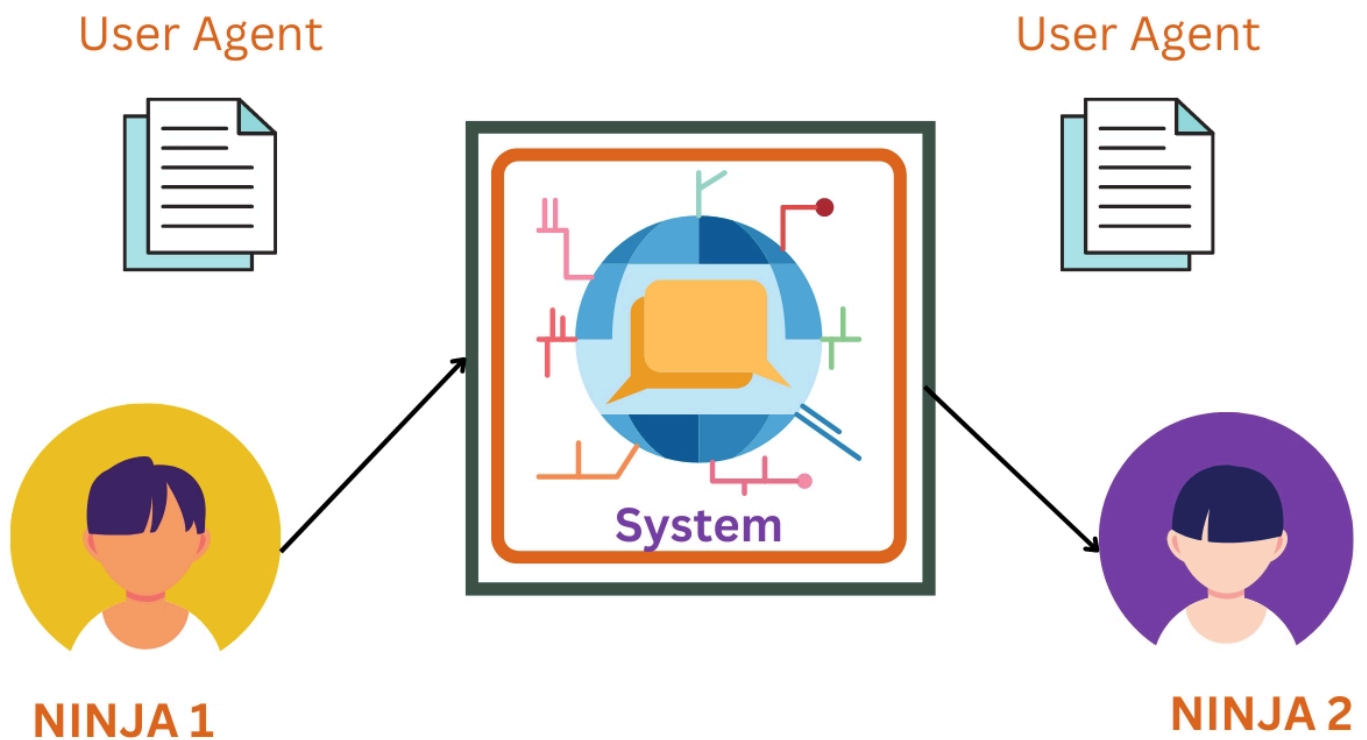
Architecture of Electronic Mail



First Scenario

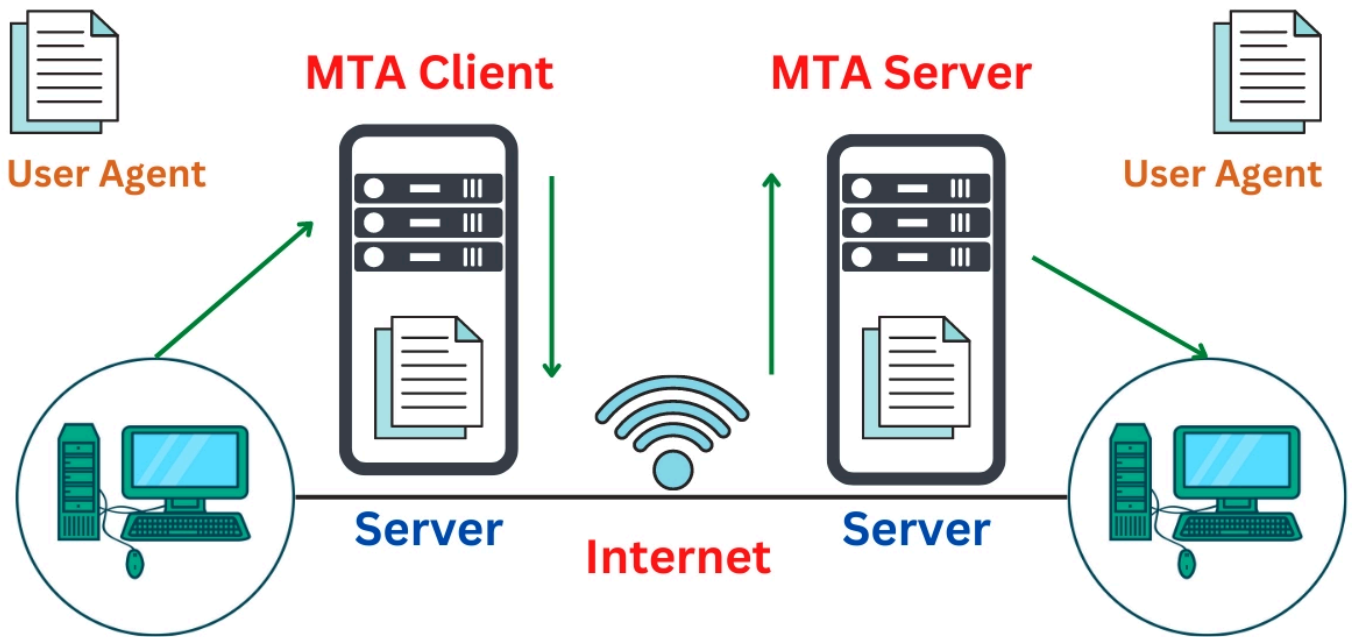
In the first scenario, two user agents are required. The sender and recipient of the e-mail share the same machine directly connected to the server.

For example, let us consider two user agents, Ninja1 and Ninja2. When Ninja1 sends an e-mail to Ninja2, the user agent (UA) programme is used to prepare the message. Following that, this e-mail gets saved in the Ninja2 inbox.



Second Scenario

In this case, the sender and recipient of an e-mail are essentially users on two different machines over the internet. User-Agents and Message Transfer Agents(MTA) are required in this scenario.

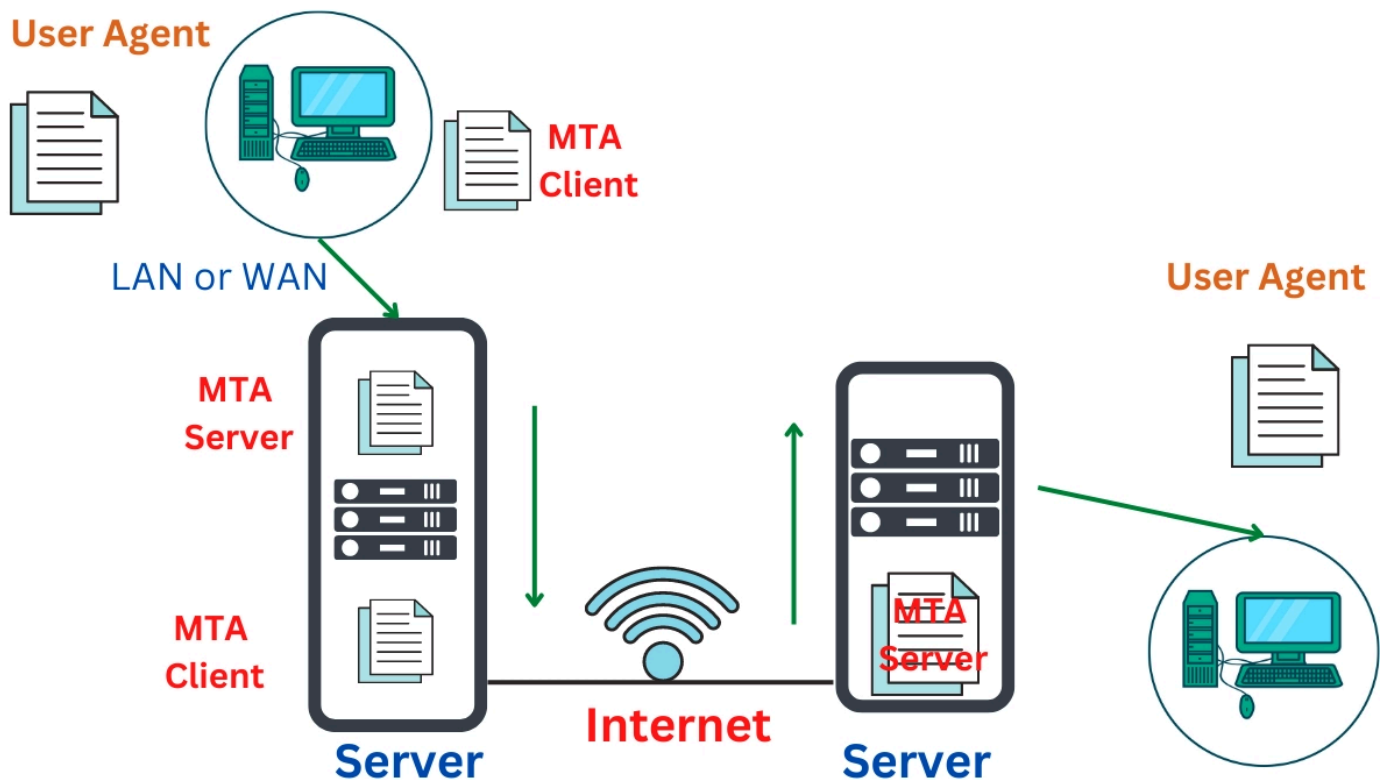


Take, for example, two user agents (Ninja1 and Ninja2), as illustrated in the diagram. When Ninja1 sends an email to Ninja2, the user agent (UA) and message transfer agents (MTAs) programmes prepare the email for transmission over the internet. Following that, this email gets stored in Ninja2's inbox.

Third Scenario

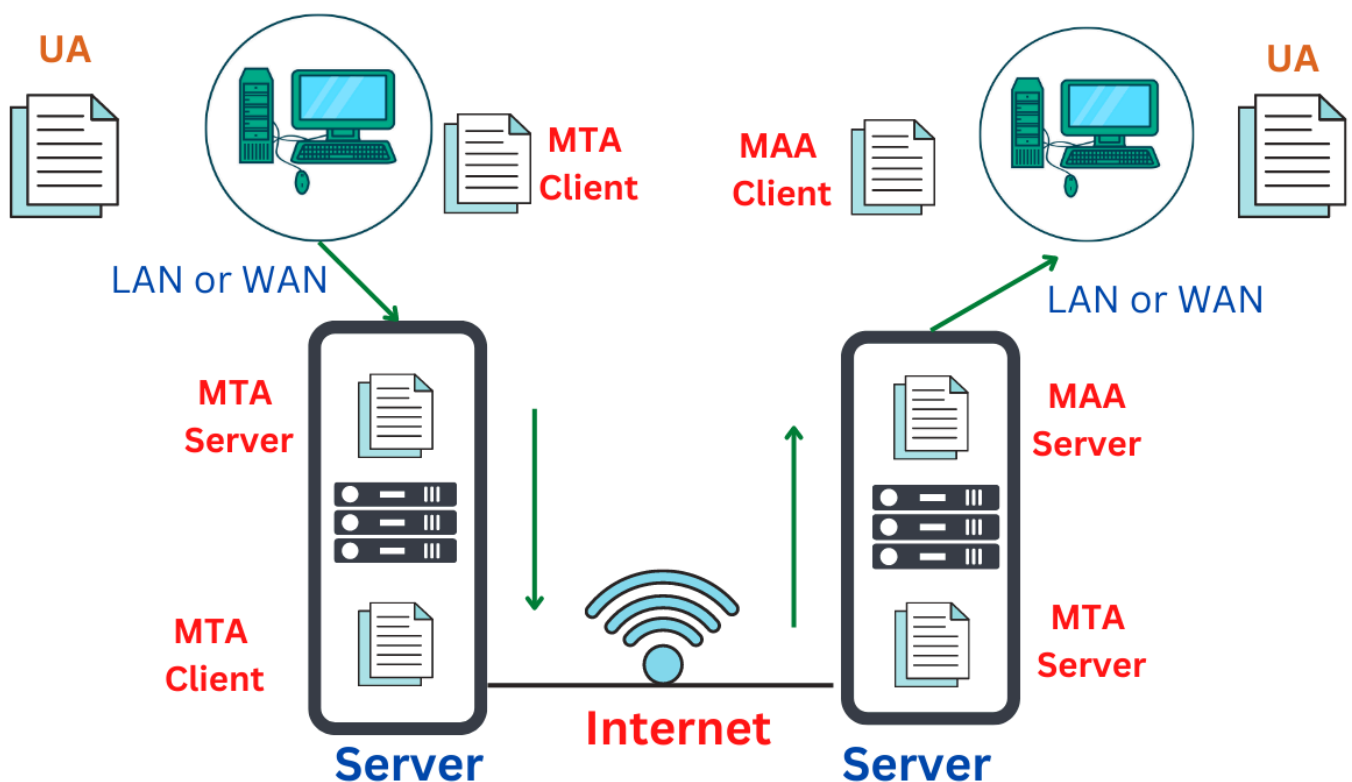
The sender is connected to the system by a point-to-point WAN, which can be a dial-up modem or a cable modem in this case. On the other hand, the receiver is directly attached to the system, as it was in the second scenario.

The sender also needs a User agent (UA) to prepare the message in this situation. After preparing the statement, the sender delivers it over LAN or WAN via a pair of MTAs.



Fourth Scenario

In this scenario, the recipient is linked to the mail server via WAN or LAN. When the message arrives, the recipient must retrieve it, which needs additional client/server agents. This scenario requires two user agents (UAs), two pairs of message transfer agents (MTAs), and a couple of message access agents (MAAs).



Different types of Email(Electronic Mail)

There are various types of email services and protocols catering to different needs and preferences. Here are some common types:

- 1. Web-Based Email:** Web-based email services are accessed through a web browser, eliminating the need for dedicated email client software. Users can access their emails from any internet-enabled device with a browser, making it a convenient and widely-used option. For example, Gmail, Yahoo Mail, Outlook.com.
- 2. Client-Based Email:** Client-based email requires dedicated email client software installed on the user's device for access. These applications offer a more feature-rich and often customizable experience compared to web-based email clients. For example, Microsoft Outlook, Mozilla Thunderbird.
- 3. Secure Email Services:** Secure email services prioritize end-to-end encryption and advanced security features to protect user privacy and sensitive information. They are often favored by individuals who prioritize secure communication. For example, ProtonMail, Tutanota.
- 4. Business or Corporate Email:** Tailored for business use, corporate email services often include collaboration tools, shared calendars, and enhanced security features to meet the specific needs of organizations. For example, Microsoft Exchange, Google Workspace.
- 5. Disposable Email Services:** Disposable email services provide temporary email addresses for short-term use. Users often utilize them for activities like online registrations or verifications, maintaining privacy. For example, Guerrilla Mail, 10 Minute Mail.
- 6. Encrypted Email Protocols:** Encrypted email protocols focus on securing the content of emails. Technologies like PGP and S/MIME employ encryption techniques to ensure confidentiality in email communication. For example, PGP (Pretty Good Privacy), S/MIME (Secure/Multipurpose Internet Mail Extensions).
- 7. POP3 (Post Office Protocol 3):** POP3 retrieves emails from the server to the local device, typically deleting them from the server. It is commonly used when users want to download and store emails locally.

World Wide Web

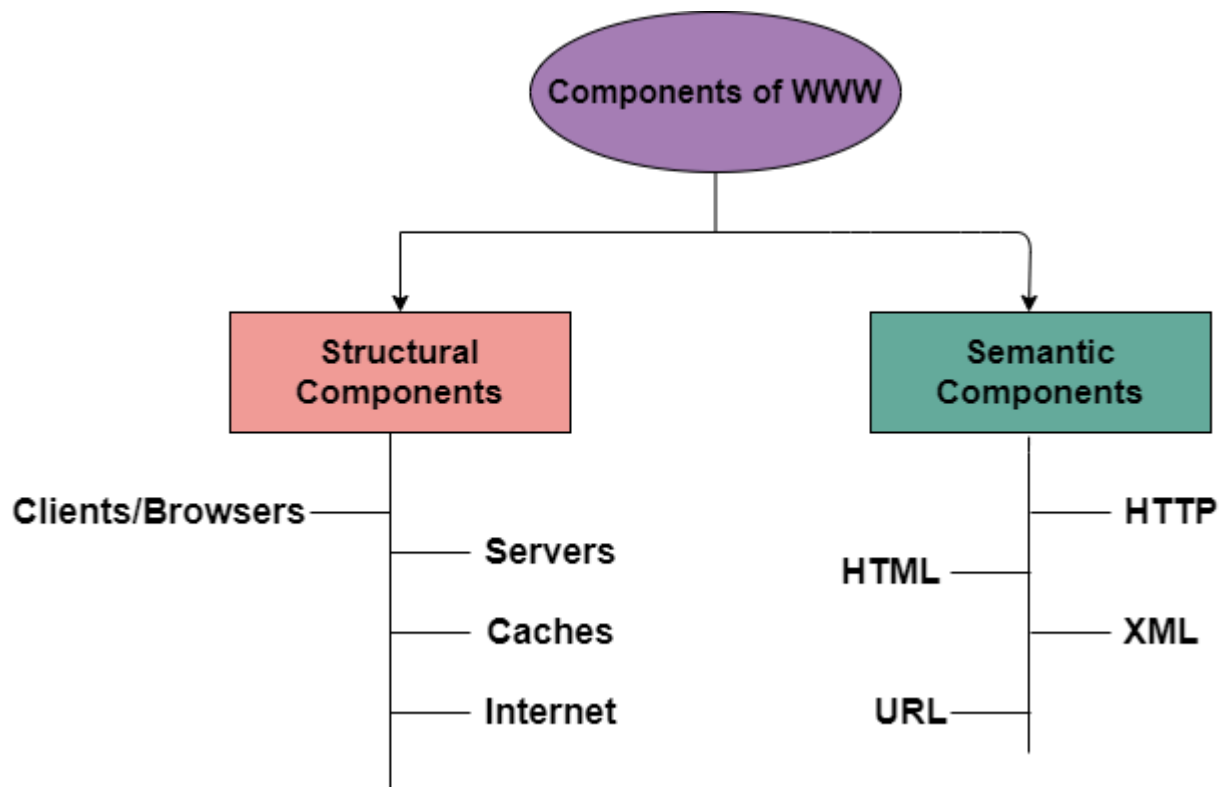
The World Wide Web or Web is basically a collection of information that is linked together from points all over the world. It is also abbreviated as WWW.

- World wide web provides flexibility, portability, and user-friendly features.
- It mainly consists of a worldwide collection of electronic documents (i.e, Web Pages).
- It is basically a way of exchanging information between computers on the Internet.
- The WWW is mainly the network of pages consists of images, text, and sounds on the Internet which can be simply viewed on the browser by using the browser software.
- It was invented by Tim Berners-Lee.

Components of WWW

The Components of WWW mainly falls into two categories:

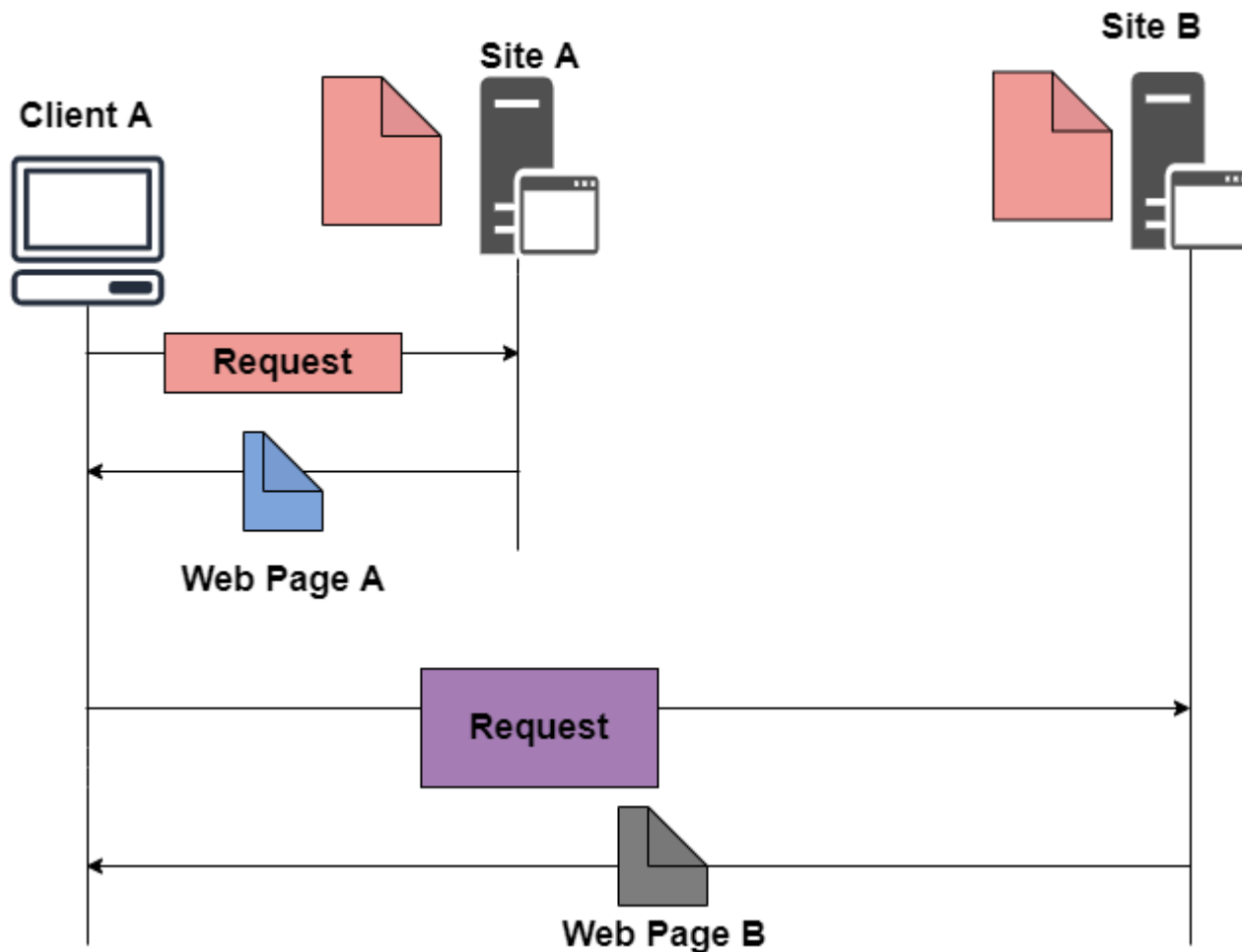
1. Structural Components
2. Semantic Components



Architecture of WWW

The WWW is mainly a distributed client/server service where a client using the browser can access the service using a server. The Service that is provided is distributed over many different locations commonly known as sites/websites.

- Each website holds one or more documents that are generally referred to as web pages.
- Where each web page contains a link to other pages on the same site or at other sites.
- These pages can be retrieved and viewed by using browsers.



In the above case, the client sends some information that belongs to site A. It generally sends a request through its browser (It is a program that is used to fetch the documents on the web).

and also the request generally contains other information like the address of the site, web page(URL).

The server at site A finds the document then sends it to the client. after that when the user or say the client finds the reference to another document that includes the web page at site B.

The reference generally contains the URL of site B. And the client is interested to take a look at this document too. Then after the client sends the request to the new site and then the new page is retrieved.

Components of WWW

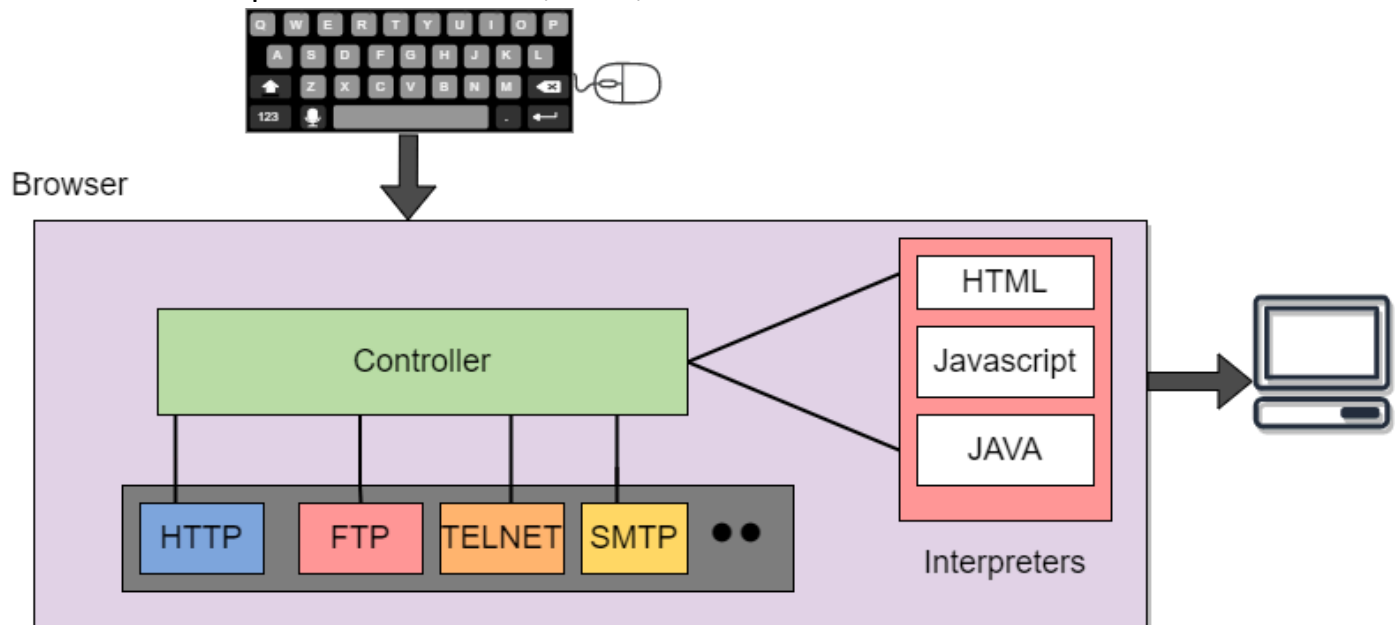
1.Client/Browser

The Client/Web browser is basically a program that is used to communicate with the webserver on the Internet.

- Each browser mainly comprises of three components and these are:
 - Controller
 - Interpreter
 - Client Protocols
- The Controller mainly receives the input from the input device, after that it uses the

client programs in order to access the documents.

- After accessing the document, the controller makes use of an interpreter in order to display the document on the screen.
- An interpreter can be Java, HTML, javascript mainly depending upon the type of the document.
- The Client protocol can be FTP, HTTP, TELNET.



2.Server

The Computer that is mainly available for the network resources and in order to provide services to the other computer upon request is generally known as the server.

- The Web pages are mainly stored on the server.
- Whenever the request of the client arrives then the corresponding document is sent to the client.
- The connection between the client and the server is TCP.
- It can become more efficient through multithreading or multiprocessing. Because in this case, the server can answer more than one request at a time.

3.URL

URL is an abbreviation of the Uniform resource locator.

- It is basically a standard used for specifying any kind of information on the Internet.
- In order to access any page the client generally needs an address.
- To facilitate the access of the documents throughout the world HTTP generally makes use of Locators.

URL mainly defines the four things:

- **Protocol**

It is a client/server program that is mainly used to retrieve the document. A commonly used protocol is HTTP.

- **Host Computer**

It is the computer on which the information is located. It is not mandatory because it is the name given to any computer that hosts the web page.

- **Port**

The URL can optionally contain the port number of the server. If the port number is

included then it is generally inserted in between the host and path and is generally separated from the host by the colon.

- **Path**

It indicates the pathname of the file where the information is located.



4.HTML

HTML is an abbreviation of Hypertext Markup Language.

- It is generally used for creating web pages.
- It is mainly used to define the contents, structure, and organization of the web page.

5.XML

XML is an abbreviation of Extensible Markup Language. It mainly helps in order to define the common syntax in the semantic web.

Features of WWW

Given below are some of the features provided by the World Wide Web:

- Provides a system for Hypertext information
- Open standards and Open source
- Distributed.
- Mainly makes the use of Web Browser in order to provide a single interface for many services.
- Dynamic
- Interactive
- Cross-Platform

Advantages of WWW

Given below are the benefits offered by WWW:

- It mainly provides all the information for Free.
- Provides rapid Interactive way of Communication.
- It is accessible from anywhere.
- It has become the Global source of media.
- It mainly facilitates the exchange of a huge volume of data.

Disadvantages of WWW

There are some drawbacks of the WWW and these are as follows;

- It is difficult to prioritize and filter some information.
- There is no guarantee of finding what one person is looking for.
- There occurs some danger in case of overload of Information.
- There is no quality control over the available data.
- There is no regulation.

HTTP

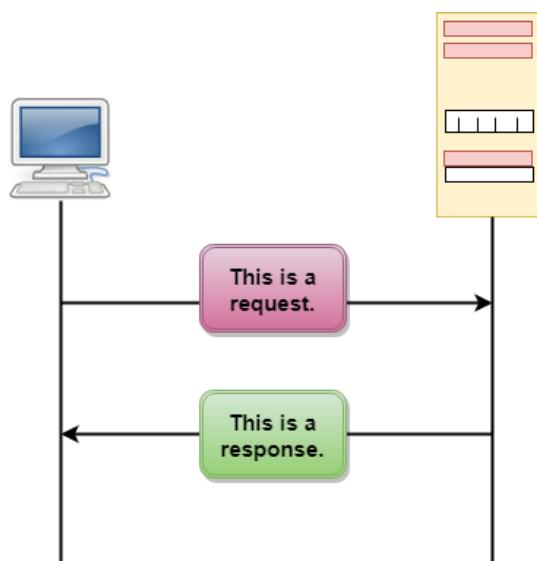
- HTTP stands for HyperText Transfer Protocol.
- It is a protocol used to access the data on the World Wide Web (www).
- The HTTP protocol can be used to transfer the data in the form of plain text, hypertext, audio, video, and so on.

- o This protocol is known as HyperText Transfer Protocol because of its efficiency that allows us to use in a hypertext environment where there are rapid jumps from one document to another document.
- o HTTP is similar to the FTP as it also transfers the files from one host to another host. But, HTTP is simpler than FTP as HTTP uses only one connection, i.e., no control connection to transfer the files.

Features of HTTP:

- o Connectionless protocol: HTTP is a connectionless protocol. HTTP client initiates a request and waits for a response from the server. When the server receives the request, the server processes the request and sends back the response to the HTTP client after which the client disconnects the connection. The connection between client and server exist only during the current request and response time only.
- o Media independent: HTTP protocol is a media independent as data can be sent as long as both the client and server know how to handle the data content. It is required for both the client and server to specify the content type in MIME-type header.
- o Stateless: HTTP is a stateless protocol as both the client and server know each other only during the current request. Due to this nature of the protocol, both the client and server do not retain the information between various requests of the web pages.

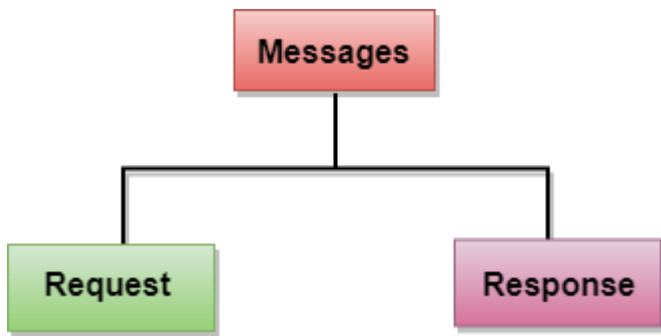
HTTP Transactions



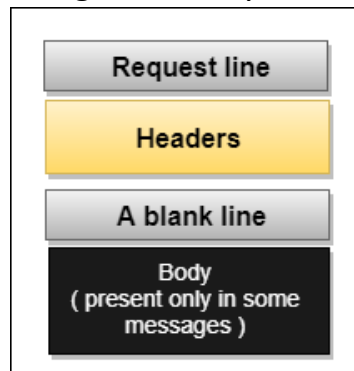
The above figure shows the HTTP transaction between client and server. The client initiates a transaction by sending a request message to the server. The server replies to the request message by sending a response message.

Messages

HTTP messages are of two types: request and response. Both the message types follow the same message format.

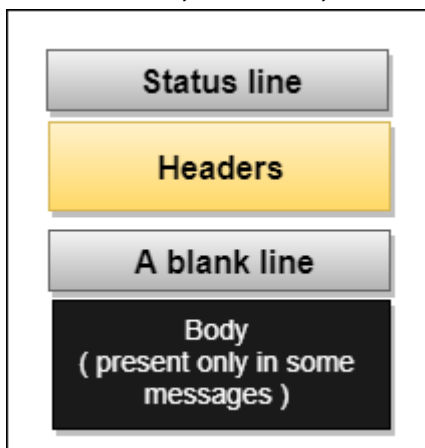


Request Message: The request message is sent by the client that consists of a request line,



headers, and sometimes a body.

Response Message: The response message is sent by the server to the client that consists of a status line, headers, and sometimes a body.



Uniform Resource Locator (URL)

- A client that wants to access the document in an internet needs an address and to facilitate the access of documents, the HTTP uses the concept of Uniform Resource Locator (URL).
- The Uniform Resource Locator (URL) is a standard way of specifying any kind of information on the internet.
- The URL defines four parts: method, host computer, port, and path.



- **Method:** The method is the protocol used to retrieve the document from a server. For example, HTTP.
- **Host:** The host is the computer where the information is stored, and the computer is given an alias name. Web pages are mainly stored in the computers and the computers are given an alias name that begins with the characters "www". This field is not mandatory.
- **Port:** The URL can also contain the port number of the server, but it's an optional field. If the port number is included, then it must come between the host and path and it should be separated from the host by a colon.
- **Path:** Path is the pathname of the file where the information is stored. The path itself contain slashes that separate the directories from the subdirectories and files.

FTP (File Transfer Protocol)

FTP stands for File Transfer Protocol, and it is basically the protocol, or procedure, for transferring files between computers. Files are not actually moved over to the source system, rather, they are copied from one computer to another. This exchange of files happens over Internet channels, formally referred to as a TCP/IP network. TCP/IP (Transmission Control Protocol/Internet Protocol) stands for the standard protocol of communication over the internet. FTP is a standard protocol, and any computer with the ability to use FTP can make use of the protocol.

CONNECTION

For FTP to work there needs to be a client and a server. The client connects to the service. The client and server communicate back and forth, the server must validate that the client has access to parts of the server it is trying to access and then the client copies the files to the directories.

Connection Modes: Active vs. Passive.

FTP may work in either active or passive mode, which sets how the data connection is setup. Each type involves the client creating a connection to the FTP server on a certain port (usually port 21). A port is a number assigned to servers in a TCP/IP network. They simply indicate the purpose of the data being transferred (e.g., web page, voice call). Servers will monitor a given port to know when data starts coming in FTP servers usually watch port 21.

Active Mode

In active mode, the client connects, indicating that it will be monitoring port 21 for incoming data from the server. Behind the scenes, the client sends a command PORT to tell

the server which port it's on. However, most clients are now behind firewalls, either personal or corporate, and the firewall doesn't let in incoming connections. For that purpose, passive mode will work better.

Active mode requires most of the setup to be completed on the client side. If using a firewall, it must be configured to allow ports for incoming connections.

Passive Mode

In passive mode, the client sends a PASV command to the server, the server then sends back an IP (Internet) address and port number. The client can use this connection to the server and data transfer/copying can occur.

For this connection mode, most of the work is on the server side. The server needs to setup the system so that it can accept incoming data, not only from port 21 but from a range of possible ports for incoming connections.

FTP uses TCP services. It needs two TCP connections. One is Control connection and another is Data connection. For control connection, it uses well-known port 21 and for data connection, it uses well-known port 20

Control Connection

A server site control connection uses a well-known port 21. There are two steps to establish a control connection –

- Server issues a passive open on the well-known port 21 and waits for the client
- After severing issues passive open, the client issue active open using an ephemeral port.

This control connection remains open throughout the process. Since the user and the server uses the interactive connection for communication, their service used by internet protocol minimizes the delay. For communication, user types the command and in return, servers give responses without any delay.

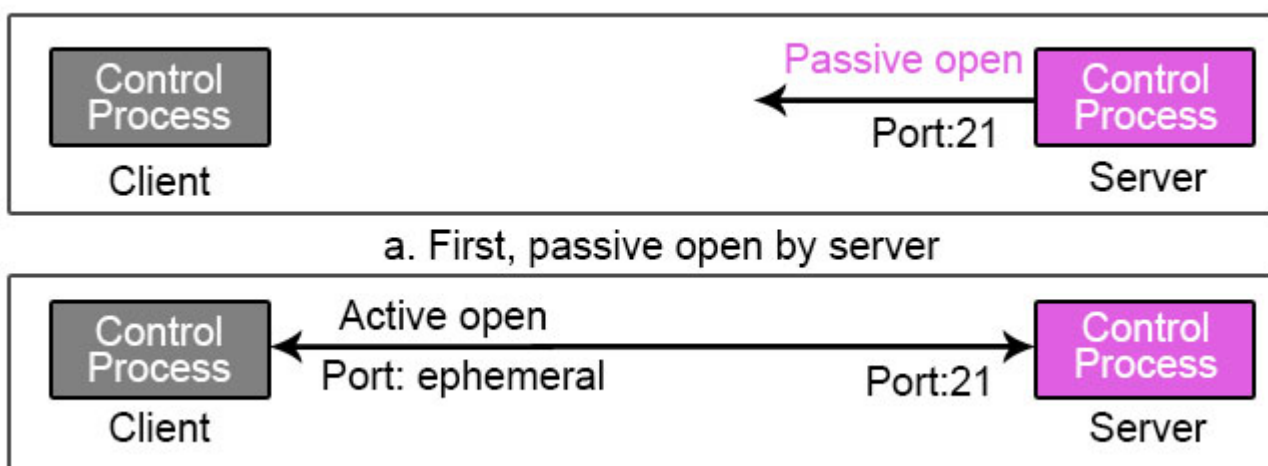


Figure 5.6: FTP Connection

Control Connection

Data Connections: At the server site, the data connection uses well-known port 20. There are three steps to establish a data connection –

- Using an ephemeral port client issues a passive open. This step must be done by the client not the server because the client wants to transform the file.
- Using the PORT command client sends this port number to the server.
- When the server receives this port number from the client, it issues active open using

well-known port 20.

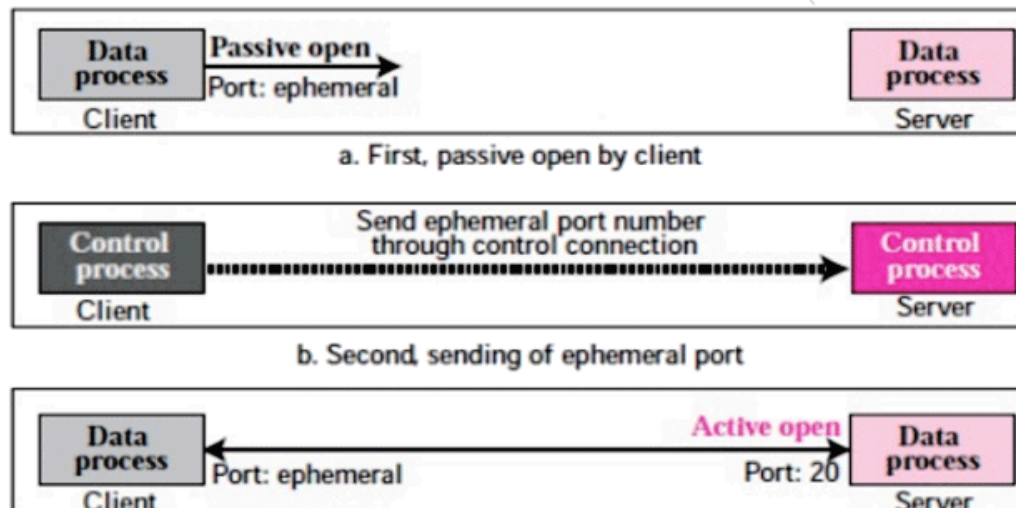


Figure 5.7: FTP Control Connection

COMMUNICATION

Both the client and the server which runs on two different systems must be communicated for transforming data. For communication, it uses two approaches i.e. communication over control connection and communication over a data connection.

Communication over the control connection

To communicate over control connection FTP uses TELNET or SMTP. Communication over control connection is done by commands and responses. The first command is sent over the connection and in return, a response is sent by another system. We can send a command or response at a time. There is only one-way communication.

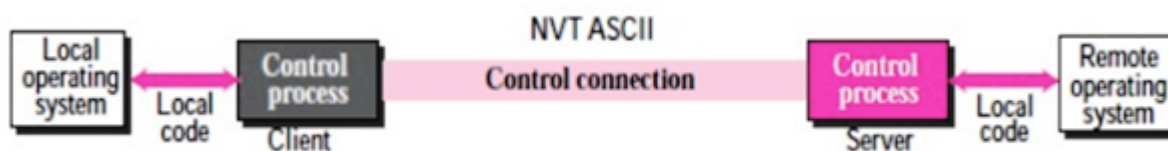


Figure 5.8: Communication over the control connection

Communication over the data connection

For transforming file over the data connection, the client must define the type of file which needs to be transformed, transmission mode, and the data structure. It solves the heterogeneity problem by defining these three attributes.

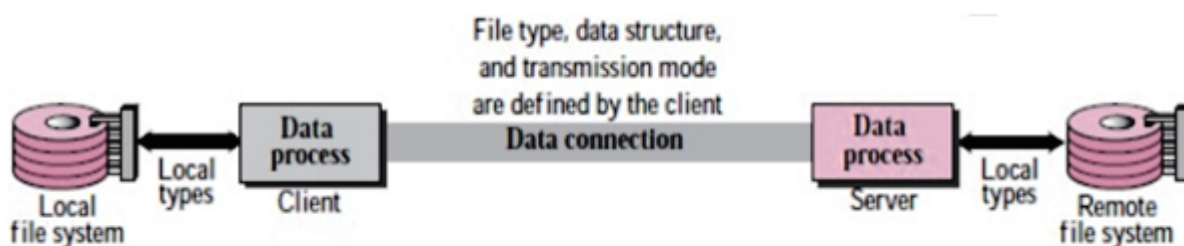


Figure 5.9: Communication over the data connection

COMMAND PROCESSING

To establish communication between the client system and the server system FTP uses a control connection. During this process, the client sends commands to the server and in return, the server sends a response to the client.

Types of FTP Transfers

It can transfer following file types across the internet connections

- 1. ASCII file:** This is the default format for transforming a file from one to another. Each character is encoded by NVT ASCII i.e. Network Virtual Terminal ASCII character set. Both the sender and the receiver transform their file from its own representation into NVT ASCII.
- 2. EBCDIC:** If sender or receiver connections use the EBCDIC encoding method, then for transforming a file FTP uses EBCDIC encoding.
- 3. Image File:** For transforming the binary file, the image file is the default mode. The file is transformed over the internet connections in the form of stream bits without encoding.

TFTP(Trivial File Transfer Protocol)

Trivial File Transfer Protocol, is a simple high-level protocol for transferring data servers use to boot diskless workstations, X-terminals, and routers by using User Data Protocol (UDP). Although it may sound similar, TFTP works differently than FTP (File Transfer Protocol) and HTTP (HyperText Transfer Protocol). Although TFTP is also based in FTP technology, TFTP is an entirely different protocol. Among the differences is that TFTP's transport protocol uses UDP which is not secure while FTP uses Transmission Control Protocol (TCP) to secure information.

TFTP was primarily designed to read or write files by using a remote server. However, TFTP is a multi-purpose protocol that can be leveraged for an array of different tasks.

TFTP Configuration Uses

TFTP configuration used for:

- Transferring files
- Remote-booting without hard drives
- Upgrading codes
- Backing up network configurations
- Backing up router configuration files
- Saving IOS images
- Booting PCs without a disk

After a workstation has been booted from a network card's ROM, the TFTP installation will download a program and then run it from a central server.